

Inception of Layout – CMOS fabrication process

Create Active regions

This section introduces the theoretical basis for the 16-mask (16M) process in semiconductor manufacturing, focusing on **substrate selection, substrate doping and resistivity, the formation of active regions**, and the critical concepts of **isolation using LOCOS (Local Oxidation of Silicon)**. Core fabrication ideas—including the use of materials like silicon dioxide, silicon nitride, and photoresist—are explained, with an emphasis on the underlying principles and reasons for each step. No implementation or procedural code is included.

Core Concepts

Substrate Selection

The fabrication process begins with choosing a **substrate**, which serves as the foundational material upon which the entire integrated circuit is built. The most common substrate in modern semiconductor devices is **P-type silicon**, characterized by its specific electrical and physical properties:

- **High resistivity:** Determines how easily electric currents can pass through the material.
- **Doping level:** Defined as the concentration of impurity atoms deliberately introduced to control the substrate's electrical behavior. For P-type silicon, a typical doping concentration is 10^{15} atoms per cubic centimeter.
- **Crystal orientation:** A critical parameter that defines the atomic arrangement on the surface, affecting further fabrication steps and device performance.

The choice of doping level ensures that the substrate's doping is lower than that used subsequently for well formation (to maintain electrical separation between wells and substrate).

Local Oxidation of Silicon (LOCOS)

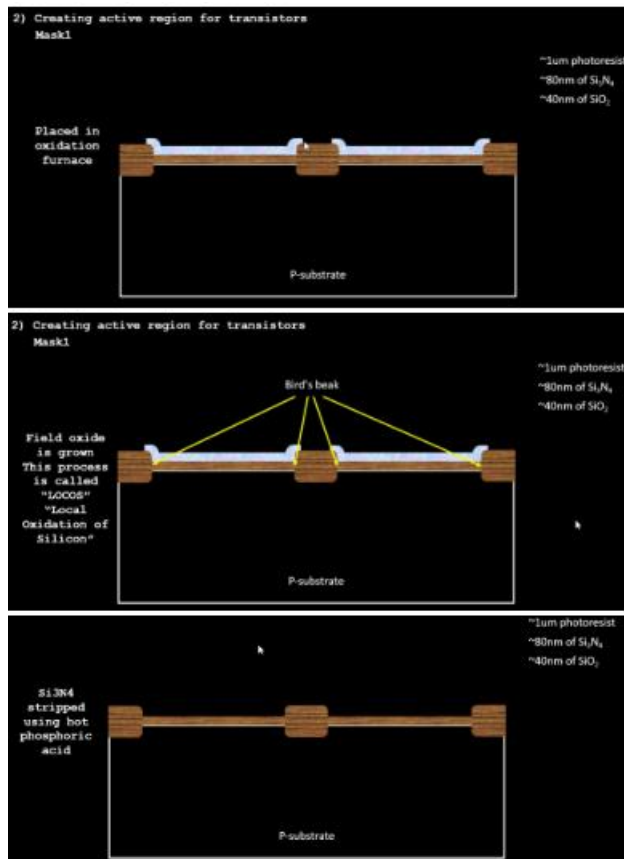
LOCOS is a critical technique for creating precise isolation between transistors:

- The wafer is exposed to high-temperature oxidation, growing a thick field oxide (SiO_2) only in unmasked areas.
- Regions under the silicon nitride mask remain protected, inhibiting oxide growth and thus defining the “active regions”.

A unique geometric effect called “**bird's beak**” occurs at the edge of growing oxides, slightly intruding under the mask.

Electrical Isolation: The resulting thick oxide physically and electrically separates neighboring transistor regions to prevent leakage or crosstalk.

- **Field Regions:** With thick oxide, function as insulators.
- **Active Regions:** Protected from oxidation, reserved for transistor fabrication.



Key Points

- The **substrate** is almost always **P-type silicon** due to its favorable electrical properties and compatibility with modern device fabrication.
- **Doping concentration** for the substrate is commonly set at 10^{15} cm^{-3} to ensure optimal separation from higher-doped well regions.
- **Photolithography** and **masking** techniques enable precise patterning of the substrate for subsequent processing.
- **Isolation** between transistors is achieved by LOCOS, which grows thick oxide in designated areas, leaving others undisturbed for device fabrication.
- The term "**bird's beak**" refers to the encroachment of the oxidation front at the boundary of the silicon nitride mask, a byproduct of the LOCOS process.

- **Successful electrical devices** critically depend on the integrity of these isolation and patterning techniques to ensure individual transistor performance and prevent undesired interaction.
 - The development of **active and isolation regions** sets the foundation for all subsequent steps in the 16M process, including well formation for PMOS and NMOS transistors.
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Formation of N-well and P-well

Summary

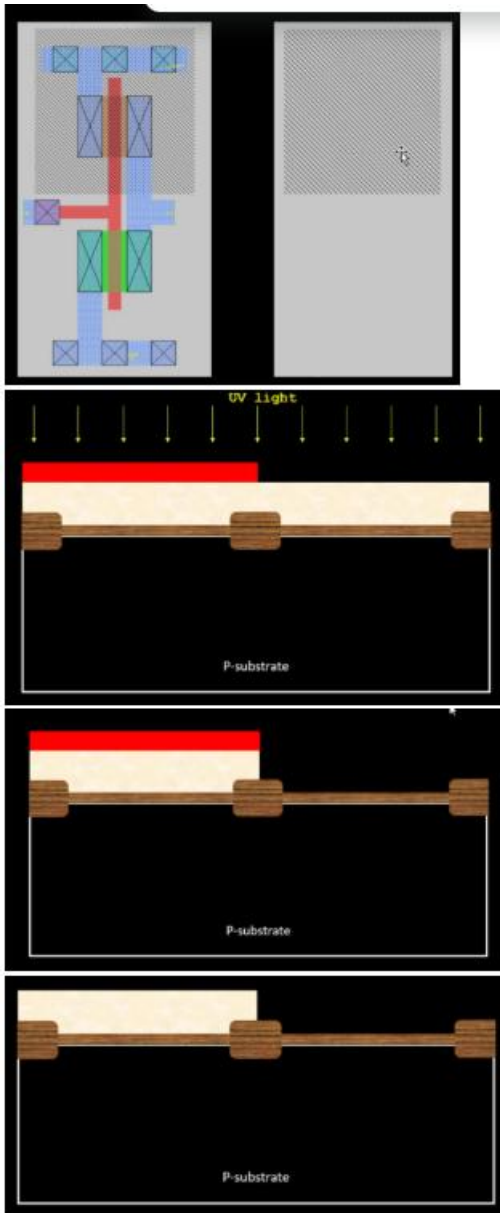
This documentation explains the **theoretical concepts** behind P well and N well (also known as "twin-tub") formation for CMOS fabrication, focusing on layout masks, photolithography, doping via ion implantation, and the role of drive-in diffusion. It emphasizes how selective processing and energy-driven doping create distinct regions for PMOS and NMOS transistors in integrated circuits.

Core Concepts

Masking and Photolithography

A **mask** is used during fabrication to selectively expose areas of a wafer. The process begins by depositing a photoresist layer, then defining patterns (using a mask) to protect or expose necessary regions. Photolithography allows precise control over which regions undergo further reactions.

- Only the areas exposed to **UV light** through the mask become chemically reactive.
- After exposure, the photoresist in the uncovered areas is removed, revealing regions for wells or active device formation.



Key Points

- **Selective Masking** is essential for separating NMOS and PMOS processes, preventing simultaneous doping in undesired areas.
- **Photoresist and UV exposure** enables controlled creation of active regions on the wafer by removing resist only from exposed areas.
- **Ion implantation energy (E)** is crucial. Boron (for P well) and phosphorus (for N well) both require sufficient kinetic energy to penetrate the oxide:
 - Typical implantation energy for boron: $E \approx 200,000$ eV
 - Phosphorus, being heavier, needs even higher E .

- **Drive-in diffusion** ensures that both wells reach the required depth and occupy substantial substrate area, making room for the full transistor structures.
- The **twin-tub approach** allows simultaneous creation of NMOS and PMOS regions, supporting complete CMOS device integration on a single chip.
- After wells are defined, subsequent steps focus on **gate formation**, which will be detailed in later documentation.

Formation of gate terminal

Summary

The gate terminal of MOS transistors (NMOS/PMOS) plays a critical role in device operation, as it directly controls the **threshold voltage**, which determines when the transistor switches 'on'. The **threshold voltage** depends on factors such as **doping concentration** and **oxide capacitance**. Precise control over these parameters during fabrication ensures robust transistor performance. The process involves steps to fine-tune doping profiles, oxide characteristics, and polysilicon gate properties, all designed to achieve the desired electrical behavior.

Core Concepts

Gate Terminal and Threshold Voltage

- The **gate** is the controlling terminal for MOS transistors. By applying a suitable voltage to the gate, one can modulate the underlying channel and thus determine whether current can flow between source and drain.
- The **threshold voltage** (V_{th}) is the minimum gate voltage required to turn the transistor 'on'. It is defined by the fundamental equation:

$$V_{th} = V_{FB} + 2\phi_F + \gamma \sqrt{2\phi_F} + V_{SB}$$

where γ (body effect coefficient) is influenced by the **substrate doping** (N_A) and **oxide capacitance** (C_{ox}).

Threshold Voltage Equation: *Snippet from "Circuit design and SPICE simulation" course*

$$V_t = V_{to} + \gamma(\sqrt{|-2\Phi_f + V_{sb}|} - \sqrt{|-2\Phi_f|})$$

Where
 V_{to} = Threshold voltage at $V_{sb} = 0$, and is a function of manufacturing process
 γ = body effect coefficient, expresses the impact of changes in body bias V_{sb} (Unit is $V^{0.5}$)
 Φ_f = Fermi Potential

$$\gamma = \frac{\sqrt{2qNA\epsilon_{si}}}{C_{ox}}$$

ϵ_{si} = relative permittivity of silicon = 11.7
 N_A = doping concentration
 q = charge of the electron
 C_{ox} = oxide capacitance

$$\Phi_f = -\Phi_T + \ln \frac{N_A}{n_i}$$

n_i = intrinsic doping parameter for the substrate

$V_{sb} = +ve$ value

Semiconductor surface inverts to n-type material at voltage $V_{gs} = V_{to} + V1$

udemy

Effect of Doping and Oxide Capacitance

- **Doping concentration** in the semiconductor substrate determines the electrical characteristics of the channel and directly affects γ .
- The **oxide capacitance** (C_{ox}) is dependent on the thickness (t_{ox}) and permittivity (ϵ_{ox}) of the gate oxide:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

- **Threshold voltage tuning** is achieved by carefully selecting the doping profile and gate oxide thickness during fabrication. Any change in these factors will impact V_{th} .

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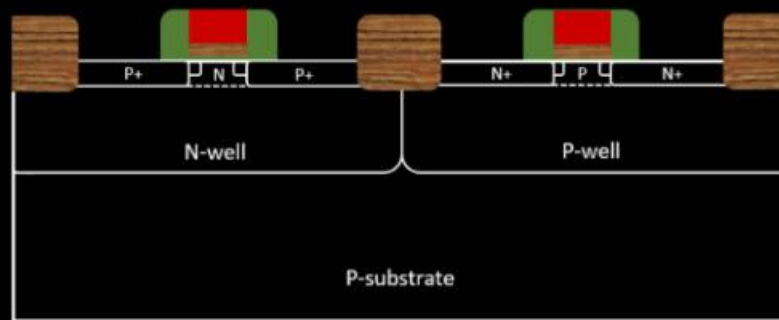
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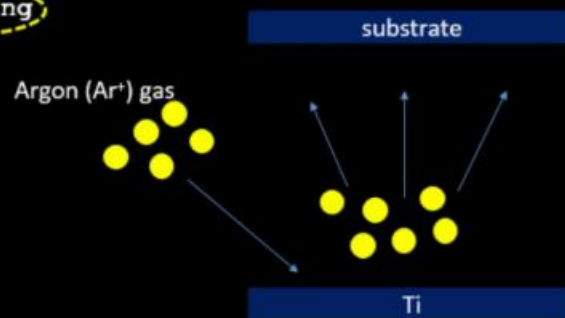
n_i = intrinsic doping parameter for the substrate

2 important terms for gate formation, as they control V_t

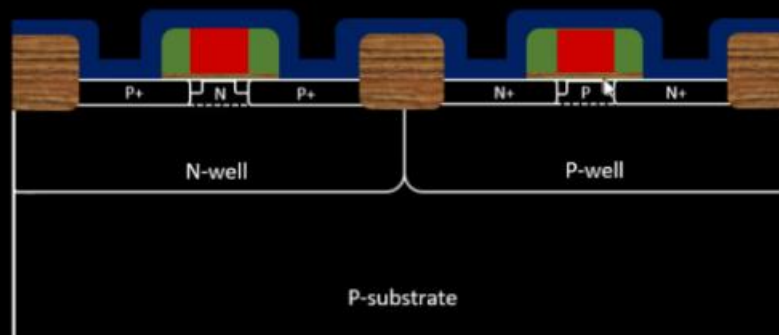
deposit titanium on wafer
surface, using sputtering

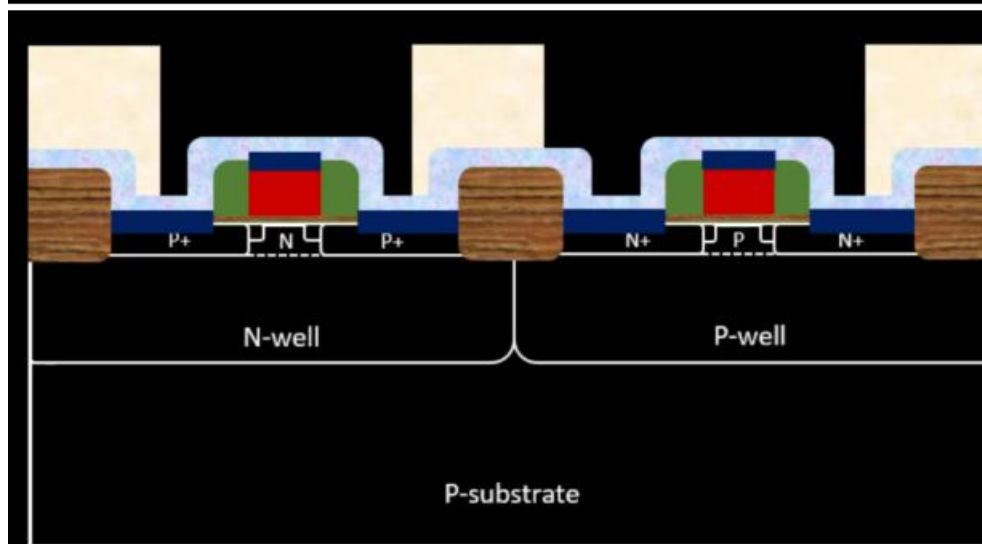
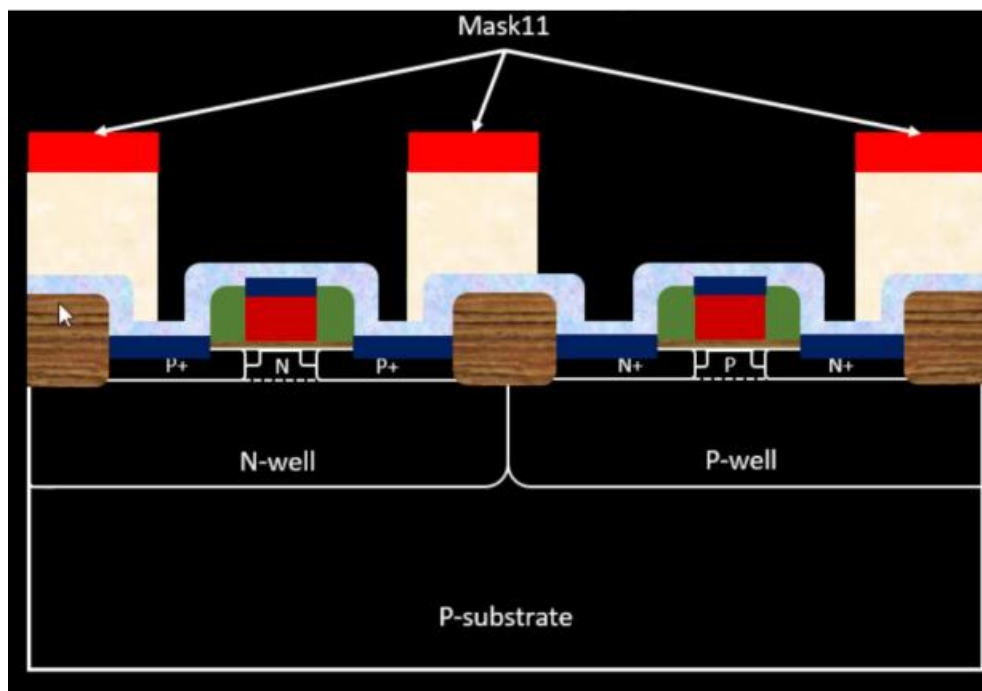


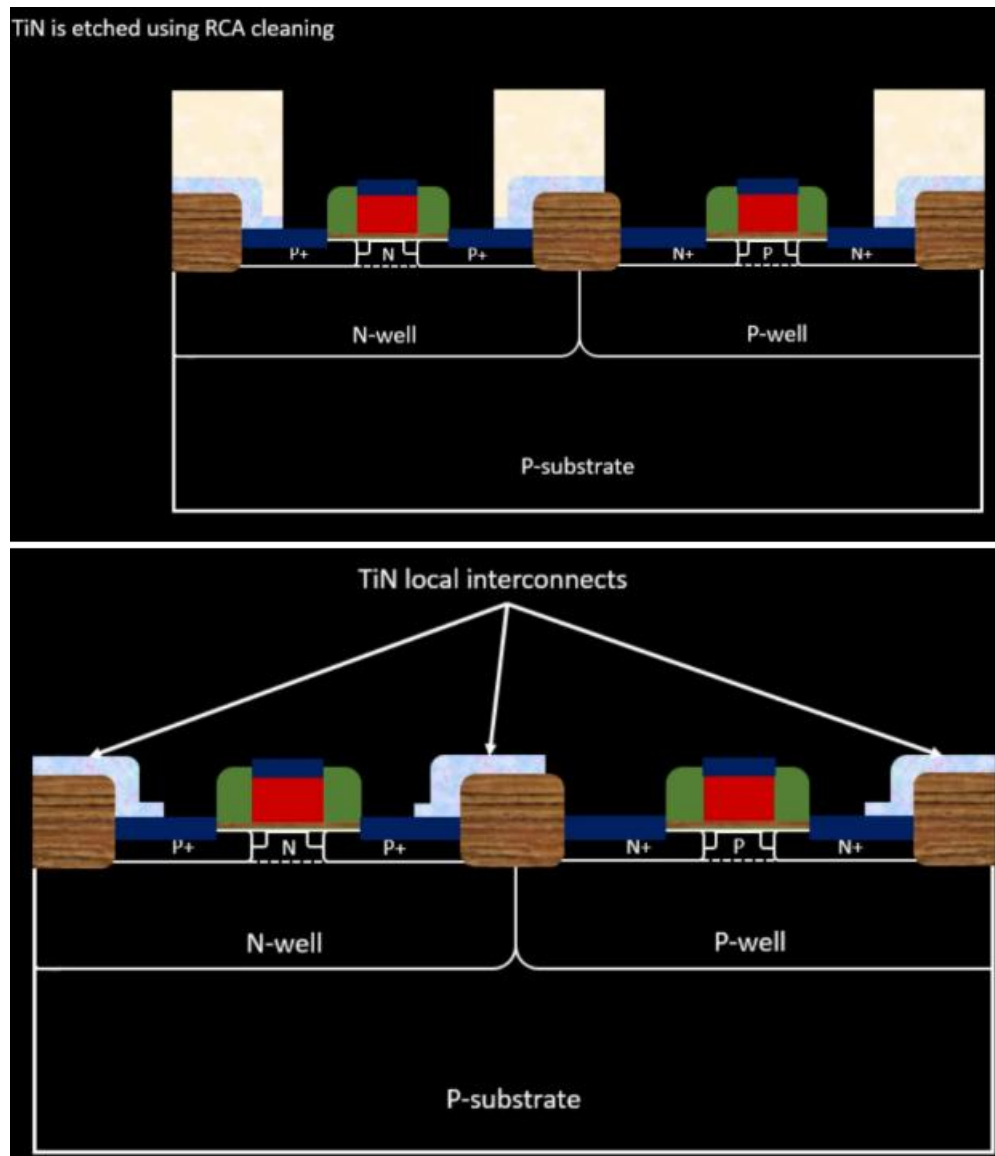
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Key Points

- **Contacts** are essential interfaces that allow electrical access to the source, drain, and gate of transistors in CMOS devices.
- **Oxide etching** with hydrofluoric acid exposes the regions where contacts will be made.
- **Sputtering** of titanium forms the initial metal layer for contacts.
- **Annealing** in nitrogen ambient at high temperatures causes titanium to react with silicon and nitrogen, forming **TiSi₂** and **TiN** respectively.
- **TiSi₂** provides low resistance pathways suitable for local wiring; **TiN** is also used for local interconnects due to its conductive properties.

- Selection of which contacts are connected locally versus routed to higher levels depends on circuit functionality (e.g., tying drains of PMOS/NMOS together or bringing outputs off-chip).
 - **Photolithography and RCA cleaning** processes ensure that only the necessary regions maintain conductive contacts after fabrication steps.
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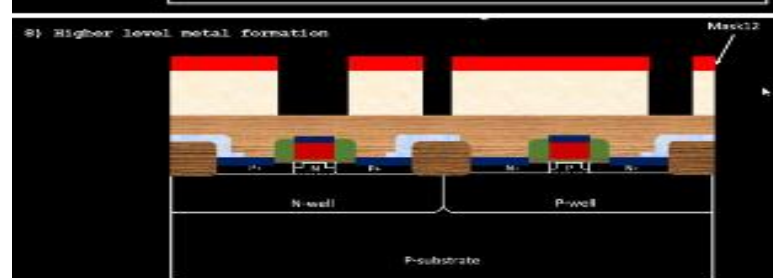
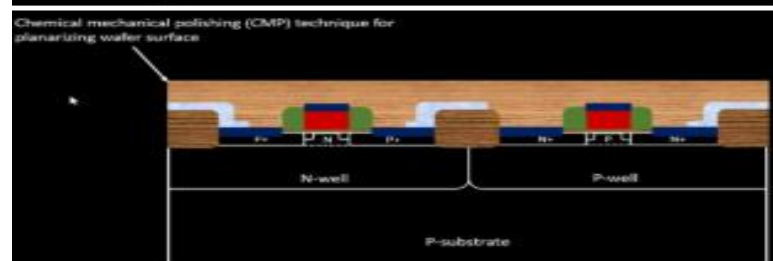
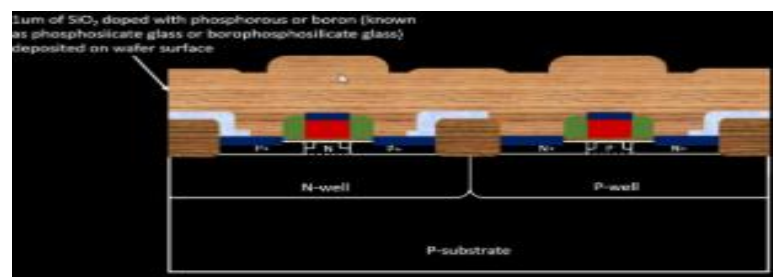
Higher level metal formation

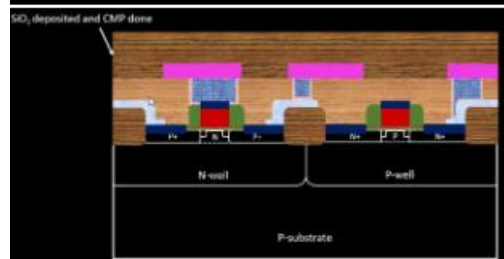
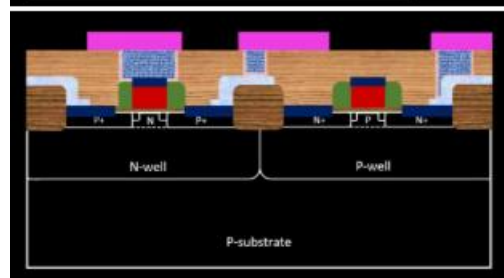
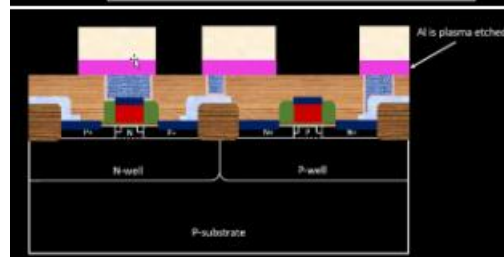
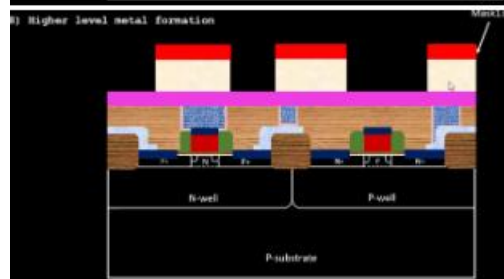
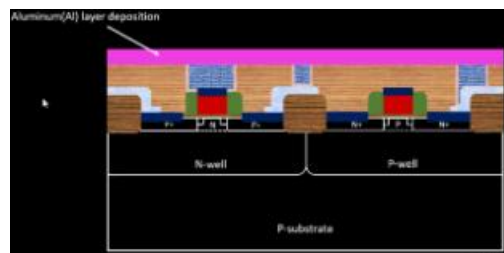
This section elaborates on the theoretical aspects of higher-level metal formation within microelectronic fabrication, focusing on the need for planarization, dielectric deposition, barrier layers, and hierarchical interconnect design. The process ensures reliable metal interconnections across multiple layers by addressing surface topography, material stability, and electrical isolation.

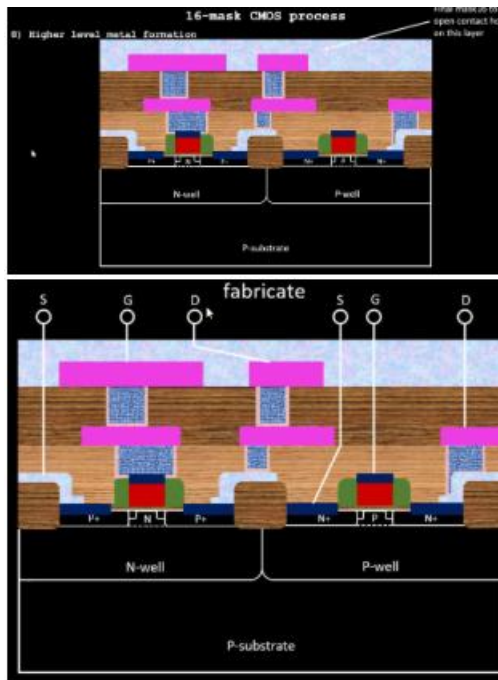
Core Concepts

Non-Planar Surface Topography & Planarization

- After initial device fabrication steps, the wafer's surface is **non-planar**, containing "hills and valleys" due to underlying features.
- Such non-flat surfaces are unsuitable for directly depositing metal interconnects, as this can cause discontinuities and reliability issues.
- To address this, a **thick SiO₂ layer** is deposited as a dielectric. This oxide may be **doped with phosphorus** (for sodium ion barrier) or **boron** (to assist in high-temperature process control).
- **Chemical Mechanical Polishing (CMP)** is used to planarize the surface, creating a uniformly flat layer upon which metal can be deposited.







Key Points

- **Planarization** is *crucial* for reliable multi-level metallization; it prevents discontinuities and supports high-density interconnects.
- **Dielectric layers** (e.g., SiO_2 , sometimes doped) provide electrical isolation and structural support between metal layers.
- **Contact holes (vias)** are formed using photolithography and are essential for interlayer connectivity; each new level requires a new mask.
- **Barrier layers** such as titanium nitride are critical for adhesion and to prevent unwanted diffusion between metal layers.
- **Metal layers** are deposited sequentially, with thickness often increasing at higher hierarchical levels to accommodate more current and lower resistance.
- **CMP** ensures each layer begins with a flat surface, maintaining process integrity through each metallization level.
- The **final passivation layer** protects the finished chip from environmental factors and provides mechanical strength.